Program: **Diploma in Food Processing Technology**	
Course Code: **6421A**	Course Title**: Fruit and Vegetable Processing**
Semester: **6**	Credits: **4**
Course Category: **Program Elective**	
Periods per week: **4(L:4 T:0 P: 0)**	Periods per semester: **60**

Course Objectives:

1. To understand the characteristics, composition, structure and maturity of fruits and vegetables
2. To study the minimal processing technologies for fruits and vegetables
3. To impart knowledge about the various fermented fruit and vegetable products

Course Prerequisites:

Topic	Program/Course Name
Basic knowledge of processing techniques	Food processing technologies

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain the principles and methods of fruit and vegetable preservation, including drying, dehydration, freezing, concentration, canning, chemical preservation, and hurdle technology.	17	Understanding
CO2	Describe the physical and textural characteristics, structure and composition, maturity standards, ripening processes, and storage practices of fruits and vegetables.	17	Understanding
CO3	Overview the processing and packaging techniques of fresh cut fruits and vegetables, vegetable wafers, soup powders, and various types of beverages such as juice, RTS, nectar, squash, concentrates, and cordials	13	Understanding

CO4	Describe the technology and methods of vinegar fermentation, fermented fruit wines, and the regulations related to fruit and vegetable products.	13	Understanding

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Stronglymapped,2-Moderatelymapped,1-Weaklymapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Explain the principles and methods of fruit and vegetable preservation, including drying, dehydration, freezing, concentration, canning, chemical preservation, and hurdle technology.**		
M1.01	Summarize on fruit and vegetable processing industry	4	Understanding
M1.02	Discuss principles of preservation-Drying – dehydration-freezing-concentration-canning-chemical preservation	8	Understanding
M1.03	Explain Hurdle technology	5	Understanding
Contents: Introduction and Scope of fruit and vegetable processing industry, overview of principles of preservation-Drying – dehydration-freezing-concentration-canning-chemical preservation, Hurdle technology and its application in fruit and vegetable industry.			
CO2	**Describe the physical and textural characteristics, structure and composition, maturity standards, ripening processes, and storage practices of fruits and vegetables.**		
M2.01	Explain physical, textural characteristics, structure and composition	7	Understanding
M2.02	Discuss about maturity of commodities, its standards, importance and methods	5	Understanding
M2.03	Summarize on fruit ripening, its chemical changes, methods and regulations	4	Understanding

	Series Test –I	1	
Contents: Fruits and vegetables-Physical characteristics, Textural characteristics, structure and composition, Maturity standards; Importance, methods of Maturity attainment, Fruit ripening- chemical changes-regulations- methods -Storage practices			
CO3	**Overview the processing and packaging techniques of fresh cut fruits and vegetables, vegetable wafers, soup powders, and various types of beverages such as juice, RTS, nectar, squash, concentrates, and cordials**		
M3.01	Discuss Minimal processing of vegetables	4	Understanding
M3.02	Explain processing of vegetable wafers, vegetable soup powder	2	Understanding
M3.03	Summarize preparations and specifications of different types of beverages	7	Understanding
Contents: Minimal processing of vegetables-processing and packaging of fresh cut fruits and vegetables. Processing technology of vegetable wafers, vegetable soup powder. Preparations and specifications of different types of beverages- Juice, RTS, Nectar, Squash, concentrates and cordials			
CO4	**Describe the technology and methods of vinegar fermentation, fermented fruit wines, and the regulations related to fruit and vegetable products**		
M4.01	Explain the technology of Vinegar fermentation	5	Understanding
M4.02	Explain the technology of fermented fruit wines	5	Understanding
M4.03	Outline the rules and regulations related to fruit and vegetable products	2	Understanding
	Series Test–II	1	
contents: Technology of Vinegar fermentation—types of vinegar-methods(slow and quick process), Technology of fermented fruit wines-champagne, neera, toddy , Rules and regulations related to fruit and vegetable products			

Text / Reference

T/R	Book Title/Author
R1	Fellows, P J. Food Processing Technology: Principles and Practice. 2nd Edition, CRC/ Woodhead, 1997. ISBN 0 8493 0887 9, ISBN 1 85573 533 4

R2	Y. H. Hui, S. Ghazala, D.M. Graham, K.D. Murrell & W.K. Nip Handbook of Vegetable Preservation and Processing Marcel Dekker (2003), ISBN : 0824743016
R3	Sivasankar, B. Food Processing and Preservation, Prentice Hall of India, 2002. ISBN 10: 8120320867ISBN 13: 9788120320864
R4	Salunke, D. K and S. S Kadam Hand Book of Fruit Science and Technology: Production, Composition, Storage and Processing. Marcel Dekker, 1995, ISBN 0 8247 9643 8
R5	Suman Bhatti, Uma Varma, Fruit and vegetable processing- Organisations and Institutions; Commonwealth Publishers, 2007, ISBN 10: 8123904045 / ISBN 13: 9788123904047

Online Resources

Sl.No	Website Link
1	https://www.appropedia.org/Toddy_and_Palm_Wine_(Practical_Action_Technical_Brief)
2	https://www.bhu.ac.in/Content/Syllabus/Syllabus_3006309920200412060036.pdf
3	https://vikaspedia.in/agriculture/post-harvest-technologies/technologies-for-agri-horti-crops/technologies-for-ripening-fruits
4	https://www.lightspeedhq.com/blog/process-making-champagne/